# The Effect of Differing Levels of Copper and Aluminum in a CuAlSn Alloy on the Strength-to-Weight Ratio and of the Alloy

## Introduction

- **Lightweight materials with high strength** are essential in modern engineering fields.
- The effects of different **copper-aluminum-tin** (CuAlSn) ratios on the **strength-to-weight ratio** are not well understood.
- Most existing studies focus on **binary alloys** rather than optimizing **ternary alloy compositions**.
- This study investigates how varying CuAlSn compositions affect **mechanical strength (specifically tensile strength).**
- **Copper** is known to enhance mechanical performance through improved atomic bonding and load transfer between phases. It strengthens the metallic lattice, leading to greater resistance to deformation under applied stress.
- The goal is to identify an **optimal alloy ratio** that maximizes the **strength-to-weight ratio**.
- **Hypothesis**: Increasing the copper content will improve the alloy's mechanical performance, resulting in higher tensile strength and strength-to-weight ratio.

## Methods

Copper, aluminum, and tin were combined in specific alloy compositions. The metals were melted at approximately 1100 degrees Celsius and poured into molds preheated to 650 degrees Celsius, then cooled. This entire process took about 70 minutes. Tensile strength was measured using a Mark-10 F305 tester, which provided maximum and final load values. Tensile strength was calculated using the maximum load and the measured cross-sectional area of each sample. The tensile strength values were then used to calculate the strength-to-weight ratio for each alloy composition.

**Independent Variable**: Alloy Composition
**Dependent Variable**: Tensile Strength/Strength-to-Weight Ratio
**Control**: Pure Copper Tensile Strength/Strength-to-Weight Ratio


Induction Furnace

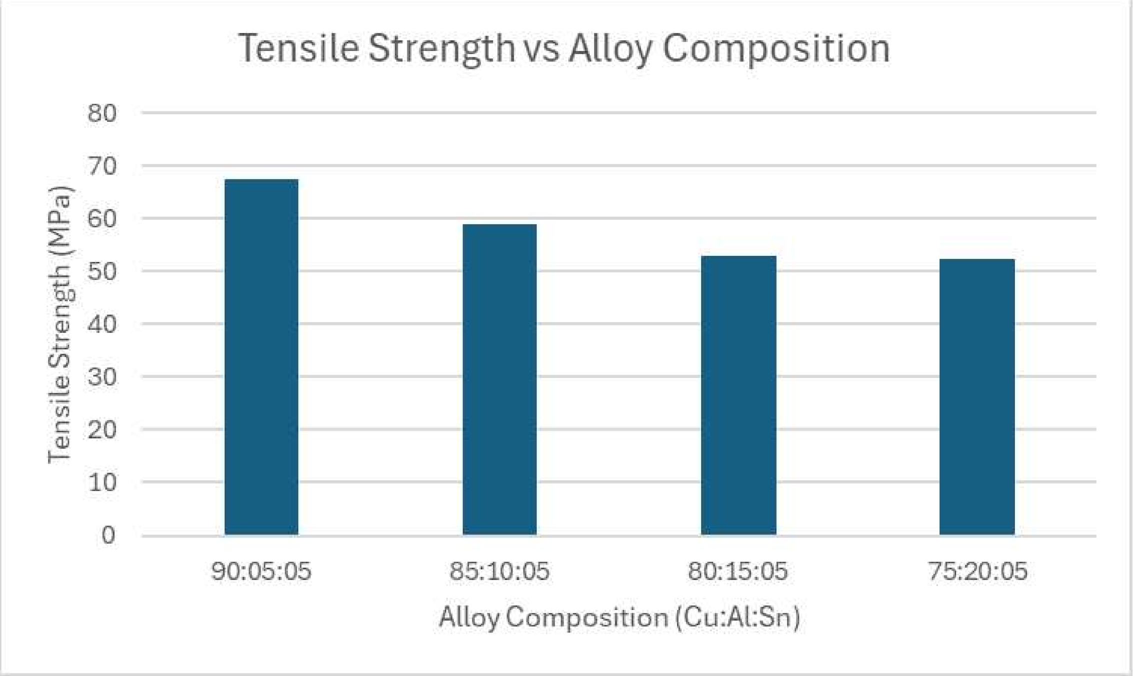

Ashing Oven

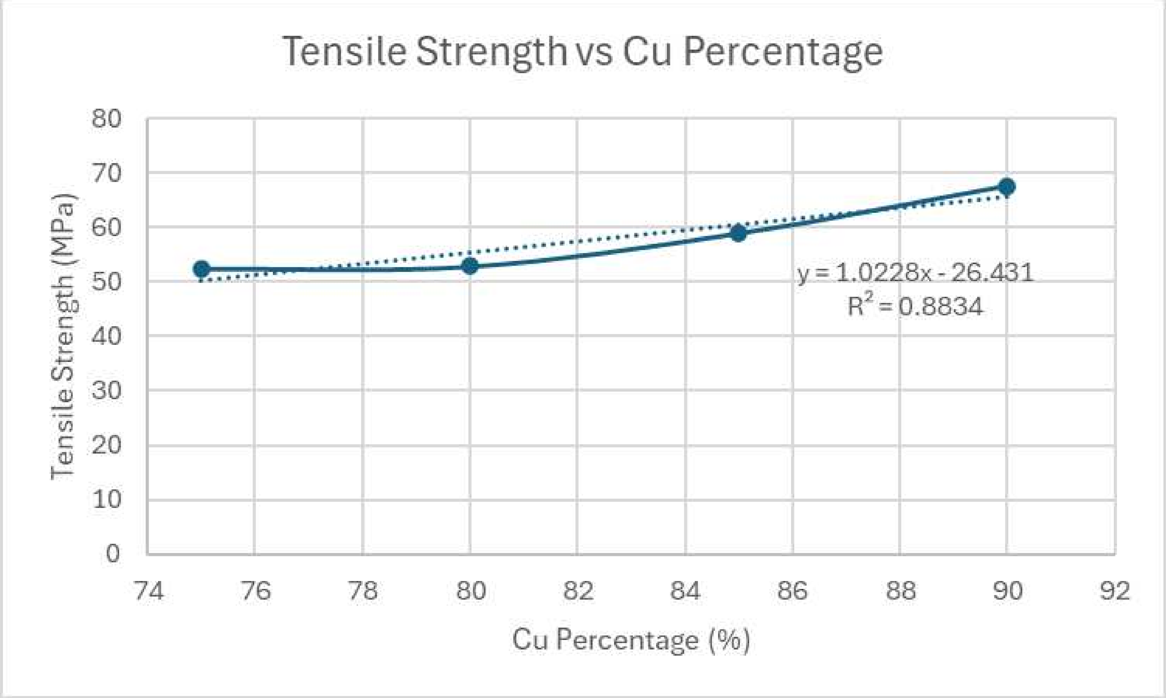
## Findings

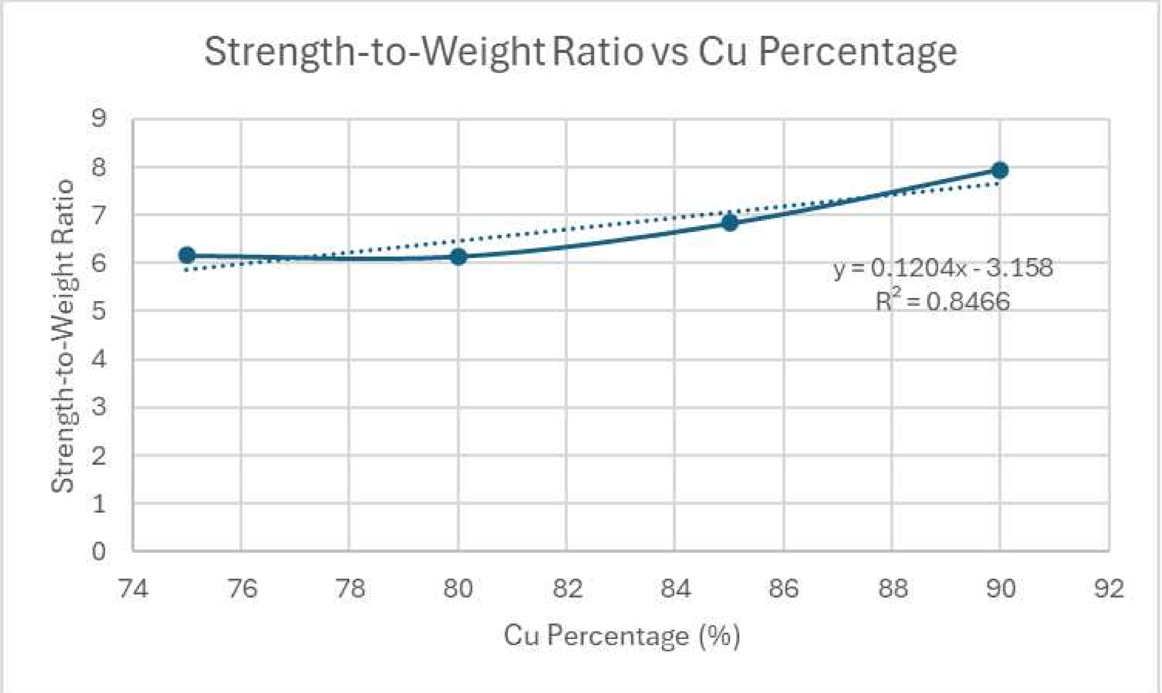
- Both tensile strength and the strength-to-weight ratio show **moderate variability** across alloy compositions.
- The spread is large enough to suggest **composition has a real mechanical effect**.
- There is a clear **positive linear trend**: As Cu increases, tensile strength rises.
- Strength-to-weight ratio increases overall with Cu%. There is a positive trend, but it's slightly less linear.
  - Small anomaly: 75% Cu and 80% Cu show nearly identical values (~6.15-6.17), which suggests a possible plateau or experimental variability.
- Increasing copper content improves mechanical performance. The relationship is approximately linear within the tested range. There is also no evidence of weakening at high Cu% (within the dataset).

Tensile Strength and Strength-to-Weight Ratio of Different Alloy Compositions

| Alloy Composition (Cu:Al:Sn) | Final Load (N) | Max Load (N) | Tensile Strength (MPa) | Strength-to-Weight Ratio |
|---|---|---|---|---|
| 90:05:05 | 1465.50 | 1485.50 | 67.52 | 7.95 |
| 85:10:05 | 1268.50 | 1296.00 | 58.91 | 6.93 |
| 80:15:05 | 1135.50 | 1163.50 | 52.89 | 6.15 |
| 75:20:05 | 1133.00 | 1154.50 | 52.48 | 6.17 |




## Discussion

Identifying how CuAlSn composition influences mechanical properties can help optimize alloy design for lightweight, high-temperature applications. Determining an optimal ratio could improve material efficiency and provide an alternative to traditional copper alloys used in demanding engineering environments.

## Limitations

The **small sample mass** and laboratory **casting conditions** may introduce variability and not fully represent industrial-scale materials. Additionally, only one sample for each ratio was tested, which **lowers precision** as there are fewer trials to compare results with and see if there were any outliers. Also, the study was mostly focused on tensile strength **rather than a full set of mechanical properties**.

## Conclusions/Future Work

- There is a **strong positive linear relationship** between Cu% and tensile strength.
- Very strong correlations observed:
  - Cu% vs Tensile Strength: r=0.94
  - Cu% vs Strength-to-Weight Ratio: r=0.93
  - Tensile Strength vs Strength-to-Weight Ratio: r≈0.999
- From the high $R^2$ and r, it can be concluded that the **copper content is the dominant predictor of mechanical performance** in this dataset.
- Strength and strength-to-weight ratio are nearly linearly dependent.
- For future work, **increasing the number of trials** per composition will improve reliability and reduce variance.
- Test a wider Cu% range to **identify possible thresholds** or saturation effects.
- Include microstructural analysis (e.g., grain size, phase distribution).
  - Helps explain why mechanical properties change with composition.
- Measure density directly to separate strength effects from weight scaling.
- Test **additional mechanical properties**: hardness, Elastic modulus, impact resistance.

## References

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